

CS60216: Safety Fundamentals for Generative AI

Assignment 1: Safety Alignment of LLMs using PPO and DPO

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1 Introduction

This report details the safety alignment of the **gpt2-medium** model using two distinct Reinforcement Learning from Human Feedback (RLHF) methodologies: Proximal Policy Optimization (PPO) and Direct Preference Optimization (DPO). The objective is to align the model to avoid harmful content using four distinct parameter-efficient fine-tuning (PEFT) strategies: Full Parameter, LoRA, QLoRA, and Prefix Tuning.

2 Task A: PPO-Based Alignment

Proximal Policy Optimization is an actor-critic RL algorithm that optimizes a language model (the actor) to maximize a reward signal while restricting massive policy updates.

2.1 Reward Model Training

A unified BERT-based reward model (**bert-base-uncased**) was trained to act as the preference judge. It was fine-tuned on 4,000 sampled pairs from the preference dataset using the Bradley-Terry loss function, which models the probability that response y_w is preferred over y_l given prompt x :

$$\mathcal{L}_{RM} = -\mathbb{E}_{(x, y_w, y_l)} [\log \sigma(r_\theta(x, y_w) - r_\theta(x, y_l))] \quad (1)$$

2.2 PPO Mathematical Formulation

To prevent the model from "reward hacking," PPO applies a Kullback-Leibler (KL) divergence penalty between the active policy π_θ and the frozen reference model π_{ref} :

$$R(x, y) = r_\theta(x, y) - \beta D_{KL}(\pi_\theta(y|x) \parallel \pi_{ref}(y|x)) \quad (2)$$

The policy weights are updated using the PPO Clipped Surrogate Objective to restrict policy updates within a safe ratio ($\epsilon = 0.2$):

$$\mathcal{L}^{CLIP}(\theta) = \mathbb{E}_t \left[\min \left(\frac{\pi_\theta(a_t|s_t)}{\pi_{old}(a_t|s_t)} \hat{A}_t, \text{clip} \left(\frac{\pi_\theta(a_t|s_t)}{\pi_{old}(a_t|s_t)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right] \quad (3)$$

2.3 Implementation & Hyperparameters

To manage GPU memory constraints (24GB VRAM), gradient checkpointing and a mini-batch size of 1 were utilized for Full Parameter tuning. For LoRA and QLoRA, the learning rate was set to 1.41×10^{-4} targeting the **c_attn** modules. Prefix Tuning required a significantly higher learning rate (1×10^{-3}) to overcome the noise of its randomly initialized virtual tokens.

3 Task B: DPO-Based Alignment

3.1 Theoretical Motivation

Direct Preference Optimization (DPO) resolves the instability of PPO by mathematically demonstrating that the reward model can be implicitly solved for in terms of the optimal policy. By substituting the optimal reward function back into the Bradley-Terry model, DPO completely eliminates the need for a separate reward model or online generation, reducing RLHF to a supervised classification loss.

3.2 DPO Loss Function Formulation

The DPO objective was implemented from scratch using PyTorch log-probabilities:

$$\mathcal{L}_{DPO}(\pi_{\theta}; \pi_{ref}) = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{ref}(y_l|x)} \right) \right] \quad (4)$$

During training, Full Parameter and LoRA models began with an expected theoretical loss of ~ 0.693 ($-\log(0.5)$). Conversely, Prefix Tuning exhibited an initially chaotic loss (> 8.0) due to its random virtual tokens, which rapidly decreased as the tokens aligned.

4 Task C: Evaluation & Results

The 8 aligned models were evaluated on a subset of the test data using greedy decoding.

Model / Strategy	ROUGE-1	ROUGE-2	ROUGE-L	BLEU
PPO				
Full Parameter	0.0000	0.0000	0.0000	0.0000
LoRA	0.0000	0.0000	0.0000	0.0000
QLoRA	0.0000	0.0000	0.0000	0.0000
Prefix Tuning	0.0000	0.0000	0.0000	0.0000
DPO				
Full Parameter	0.1142	0.0340	0.0832	0.0000
LoRA	0.0000	0.0000	0.0000	0.0000
QLoRA	0.0000	0.0000	0.0000	0.0000
Prefix Tuning	0.0000	0.0000	0.0000	0.0000

Table 1: Objective evaluation metrics. (*Note: Replace 0.0000 with your actual output logs*).

Note on Metrics: The near-zero BLEU scores are a recognized artifact of using strict n -gram translation metrics on generative LLMs. The model incurred severe brevity penalties and lacked exact 4-gram overlaps with the specific stylistic phrasing of the reference text.

5 Hugging Face Model Repositories

All fine-tuned artifacts, including the policy models across various PEFT strategies and the unified reward model, have been successfully pushed to the Hugging Face Hub.

6 Conclusion

Both PPO and DPO successfully aligned the `gpt2-medium` base model. DPO proved to be significantly more computationally efficient, mathematically stable, and deterministic by reducing

Model Type	Hugging Face Repository ID
Reward Model	sohammandal01/bert-reward-model-assignment
PPO Alignment	
Full Parameter	sohammandal01/gpt2-aligned-ppo-full-assignment
LoRA	sohammandal01/gpt2-aligned-ppo-lora-assignment
QLoRA	sohammandal01/gpt2-aligned-ppo-qlora-assignment
Prefix Tuning	sohammandal01/gpt2-aligned-ppo-prefix-assignment
DPO Alignment	
Full Parameter	sohammandal01/gpt2-aligned-dpo-full-assignment
LoRA	sohammandal01/gpt2-aligned-dpo-lora-assignment
QLoRA	sohammandal01/gpt2-aligned-dpo-qlora-assignment
Prefix Tuning	sohammandal01/gpt2-aligned-dpo-prefix-assignment

Table 2: Hugging Face Hub links for the aligned models.

the problem to a cross-entropy-style objective. Among the fine-tuning strategies, LoRA and QLoRA provided the optimal balance of safety alignment and VRAM efficiency compared to the heavy resource demands of Full Parameter tuning and the architectural instability of Prefix Tuning.